

LYUBASHIVKA, Ukraine

WMO: 337610

Lat: 47.8508N

Lon: 30.2683E

Elev: 183

StdP: 99.15

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.9	-13.8	-20.0	0.7	-16.0	-17.2	0.8	-13.1	10.4	-6.3	9.5	-5.1	2.9	0	0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.5	20.2	30.7	19.4	29.1	18.9	21.7	29.0	20.8	28.0	20.1	26.9	3.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.4	14.5	24.5	18.5	13.7	23.6	17.7	13.0	22.8	64.2	29.2	61.0	28.1	58.3	26.8	
(p)															

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.5	7.6	6.4	-19.4	34.5	3.6	2.1	-22.0	36.0	-24.1	37.2	-26.1	38.3	-28.7	39.9
			-19.7	24.0	3.6	2.3	-22.2	25.7	-24.3	27.1	-26.3	28.4	-28.9	30.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	-3.6	-1.8	3.1	10.2	16.0	19.8	21.9	21.4	15.8	9.4	4.1	-1.4
	DBStd	10.00	5.38	5.55	4.72	3.93	3.88	3.25	2.98	3.35	3.97	4.49	4.84	5.32
	HDD10.0	1623	423	329	216	44	3	0	0	0	4	66	184	353
	HDD18.3	3502	681	563	472	246	93	22	5	9	94	278	426	612
	CDD10.0	1489	0	0	3	49	189	295	367	352	178	47	8	0
	CDD18.3	326	0	0	0	1	21	67	114	103	19	1	0	0
	CDH23.3	3129	0	0	0	15	223	589	1064	1040	190	8	0	0
	CDH26.7	1045	0	0	0	3	49	166	379	404	44	0	0	0
(14) Wind	WSAvg	3.0	3.3	3.5	3.7	3.4	2.9	2.6	2.5	2.6	2.7	2.7	3.2	3.3
(15) Precipitation	PrecAvg	526	33	27	29	35	51	72	68	45	47	41	41	36
	PrecMax	669	87	65	60	74	119	176	174	109	134	121	105	105
	PrecMin	392	6	3	3	7	7	16	18	8	0	1	0	1
	PrecStd	85	20	15	16	17	29	34	44	29	37	28	25	27
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	11.9	18.1	25.3	30.5	32.2	34.6	35.1	30.4	24.3	17.8	10.9
		MCWB	5.6	7.7	10.2	15.1	17.5	19.9	20.6	20.8	18.0	15.9	12.9	9.2
	2%	DB	6.0	8.5	14.8	21.8	28.0	30.2	32.4	32.7	27.7	21.4	14.4	9.0
		MCWB	4.5	6.0	8.6	13.0	17.0	19.1	20.5	20.1	16.7	14.4	11.6	7.7
	5%	DB	4.1	6.4	12.3	19.6	26.0	28.6	30.5	30.8	25.5	18.5	12.4	7.2
		MCWB	3.2	4.8	7.1	11.7	16.3	18.9	19.9	19.5	16.1	13.3	10.8	6.3
	10%	DB	2.8	4.7	10.1	17.5	23.7	26.9	28.7	28.8	23.2	16.0	10.8	5.4
		MCWB	2.0	3.5	5.9	10.7	15.4	18.3	19.4	18.7	15.2	12.6	9.5	4.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	8.5	10.9	16.1	19.6	22.5	22.8	23.1	19.9	17.2	13.6	9.7
		MCDB	7.5	10.9	16.5	24.8	26.7	28.3	29.6	31.1	27.1	21.8	15.9	10.4
	2%	WB	4.7	6.3	9.2	13.9	18.2	21.0	21.7	21.7	18.3	15.6	12.2	7.8
		MCDB	5.8	8.0	13.6	20.1	25.0	27.3	29.0	29.8	24.5	19.4	13.8	8.5
	5%	WB	3.3	4.9	7.8	12.7	17.2	20.1	20.9	20.5	17.2	14.2	11.0	6.3
		MCDB	4.0	6.1	11.6	17.7	24.0	26.2	28.2	28.0	22.8	17.6	12.3	7.1
	10%	WB	2.1	3.6	6.4	11.4	16.2	19.2	20.2	19.6	16.1	12.8	9.5	4.7
		MCDB	2.7	4.7	9.3	16.2	22.2	24.9	27.1	26.6	21.5	15.8	10.7	5.3
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.5	7.8	10.3	10.9	10.6	11.0	11.4	10.1	8.3	5.5	4.7
		MCDBR	5.1	7.8	11.8	13.7	13.7	12.8	13.4	13.9	13.6	12.0	7.7	5.8
		MCWBR	4.2	5.5	6.8	6.8	5.7	5.0	4.7	5.2	5.8	6.2	5.0	4.7
	5% WB	MCDBR	4.8	7.2	11.0	11.9	12.2	11.2	11.8	12.2	11.5	10.2	7.0	5.4
		MCWBR	4.1	5.4	6.8	6.3	5.6	5.2	4.9	5.3	5.7	5.8	4.9	4.7
(40) Clear-Sky Solar Irradiance	taub		0.319	0.350	0.382	0.435	0.409	0.403	0.433	0.432	0.398	0.367	0.346	0.320
	taud		2.258	2.226	2.240	2.126	2.276	2.310	2.237	2.233	2.298	2.368	2.353	2.320
	Ebn at Noon		734	792	827	815	855	861	827	809	802	768	701	686
	Edh at Noon		78	101	116	144	128	125	133	128	109	87	72	66
(44) All-Sky Solar Radiation	RadAvg		1.02	1.83	3.06	4.55	5.84	6.23	6.12	5.50	3.79	2.35	1.06	0.77
	RadStd		0.17	0.30	0.37	0.41	0.50	0.42	0.37	0.31	0.51	0.29	0.17	0.17

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.62	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+0.68	N/A	N/A	+0.83	N/A	N/A	-123	-179	+117	+63

Nomenclature: See separate page